

Human Review: Very Smooth Operator Converse Counterexample

Original automatic proof: `solution_packet.pdf`

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Status: Verified.

Verdict: The proof is correct. The example does show that the converse question after Corollary 14 in Rao's paper is not true in general.

Human Comments

The proposed counterexample is elementary and convincing. Let

$$X = \ell_1, \quad Y = \mathbb{K},$$

and let $T \in \mathcal{K}(\ell_1, \mathbb{K})$ be the scalar-valued operator represented by the sequence

$$a = (a_n) \in \ell_\infty, \quad a_1 = 1, \quad a_n = 1 - \frac{1}{n} \quad (n \geq 2).$$

Since the range is one-dimensional, T is compact. Also

$$\|T\| = \|a\|_\infty = 1.$$

The operator T attains its norm only at e_1 , up to scalar multiples. Indeed, if $\|x\|_1 = 1$ and $|Tx| = 1$, then

$$1 = |Tx| \leq \sum_{n=1}^{\infty} a_n |x_n| \leq \sum_{n=1}^{\infty} |x_n| = 1.$$

Since $a_n < 1$ for every $n \geq 2$, equality forces $x_n = 0$ for all $n \geq 2$. Hence $x = \lambda e_1$, with $|\lambda| = 1$. Thus the norm-attaining vector is unique up to scalars.

Moreover,

$$T(e_1) = 1 \in \mathbb{K},$$

and every unit vector in the one-dimensional space $\mathbb{K}$ is very smooth. Therefore the hypotheses suggested by Rao's converse question are satisfied.

However, T is not a very smooth point of $\mathcal{K}(\ell_1, \mathbb{K})$. In fact,

$$\mathcal{K}(\ell_1, \mathbb{K}) = \mathcal{L}(\ell_1, \mathbb{K}) = (\ell_1)^* = \ell_\infty$$

isometrically, and T corresponds to the sequence a . This sequence is not a smooth point of ℓ_∞ . The coordinate functional

$$\delta_1(b) = b_1$$

has norm one and satisfies

$$\delta_1(a) = 1.$$

On the other hand, if $\mathcal{U}$ is a free ultrafilter on $\mathbb{N}$, then

$$L_{\mathcal{U}}(b) = \lim_{\mathcal{U}} b_n$$

defines a norm-one functional on ℓ_∞ , and since $a_n \rightarrow 1$, one has

$$L_{\mathcal{U}}(a) = 1.$$

The two norming functionals are distinct; for instance,

$$\delta_1(e_1) = 1, \quad L_{\mathcal{U}}(e_1) = 0.$$

Thus a is not a smooth point of ℓ_∞ . Consequently T is not even smooth as a point of $\mathcal{K}(\ell_1, \mathbb{K})$, and therefore it cannot be very smooth.

Short Version

The example satisfies the proposed converse hypotheses: T is compact, it attains its norm at a unique unit vector up to scalars, and $T(e_1)$ is very smooth in the scalar range. Nevertheless, T corresponds to a nonsmooth point of ℓ_∞ , because the sequence

$$(1, 1/2, 2/3, 3/4, \dots)$$

is normed both by first-coordinate evaluation and by free-ultrafilter limits. Hence T is not very smooth.

Review Outcome

This should be classified as a correct counterexample. It proves that the converse question after Corollary 14 is false in general.